Do Riots Drive Identity Underground?

Tables and figures with the frozen name index, riot years 1970–1995

19 October 2026

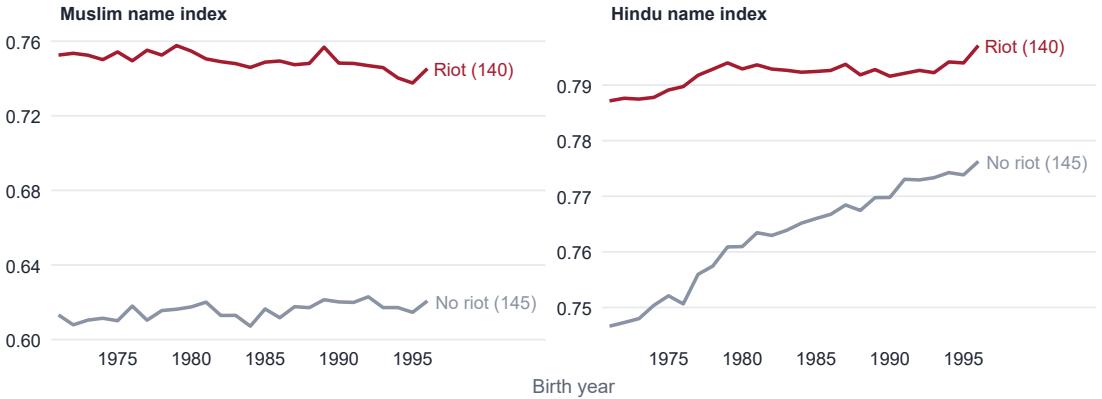
Effect of riot exposure on names

	Muslim Name Index (t+1)	Hindu Name Index (t+1)
	(1)	(2)
Panel A: OLS Estimates		
Riot exposure	0.0068***	0.0012**
	(0.0010)	(0.0005)
Panel B: Reduced Form Estimates		
Friday Festival	0.0015***	0.0014***
	(0.0006)	(0.0002)
Panel C: IV Estimates		
Riot exposure (instrumented)	0.0233***	0.0215***
	(0.0085)	(0.0033)
First-stage F-statistic	435.48	435.48
Observations	12,161	12,161
Controls	Yes	Yes
Quadratic Year Trend	Yes	Yes
District FE	Yes	Yes

Notes: Outcome: name index of the cohort born in t+1; higher values mean more distinctive names. Riot exposure is $\exp(-d^2/(2 \cdot 200^2))$, where d is the distance in km to the nearest district with a riot that year; Panel C instruments it with the Friday festival (Iyer and Shrivastava 2018). Controls: BJP rule, Muslim share, urbanisation, literacy, electricity and tap water. Standard errors clustered by district in parentheses; the appendix adds standard errors clustered by state-year. With one instrument, the reduced-form test is the Anderson–Rubin test of a zero effect, which stays valid when the instrument is weak. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

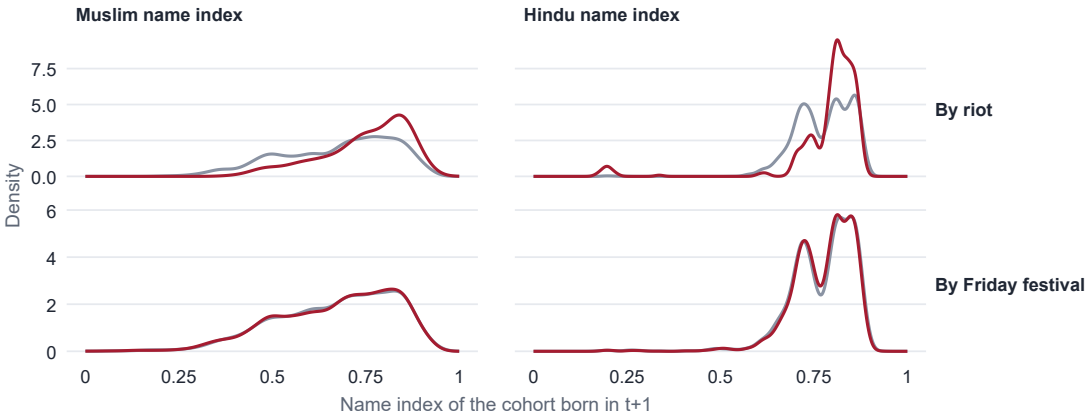
Name index by birth cohort

Mean name index of each birth cohort in districts on the 1981 list. Crimson: districts with at least one riot in 1970–1995; grey: districts with none. Numbers of districts in brackets.



Distribution of the name index

Density of the name index of the cohort born in t+1. Top row: district-years with a riot (crimson) and without (grey), districts on the 1981 list. Bottom row: with a Friday festival (crimson) and without (grey), all districts.



Appendix

Summary statistics

	Panel A: Full Sample					
Variable	N	Mean	SD	Min	Max	Median
Outcome Variables						
Muslim Name Index (t+1)	12,161	0.671	0.157	0	0.949	0.698
Hindu Name Index (t+1)	12,161	0.775	0.084	0.171	0.933	0.795
Number of Muslims (scored)	12,161	2,690	3,558	1	43,602	1,488
Number of Hindus (scored)	12,161	15,033	13,308	9	105,519	11,630
Riots (Districts on the 1981 List)						
Riot occurrence	7,410	0.058	0.234	0	1	0
Number of riots	7,410	0.10	0.59	0	20	0
Deaths in riots	7,410	0.66	10.38	0	564	0
Injuries in riots	7,410	1.93	23.76	0	1,023	0
Riot Exposure & Instrument						
Riot exposure	12,161	0.496	0.345	0	1	0.522
Friday Festival	12,161	0.578	0.494	0	1	1

Notes: Name indices are the mean name score of the cohort born in t+1, with names scored on births before 1970; higher values mean more distinctive names, and the Hindu index is one minus the score. Numbers of Muslims and Hindus count voters in polling-booth cells with a scored name. Riot rows cover districts on the 1981 list, the only ones with own-district riot data; missing deaths and injuries count as 0. Riot exposure is $\exp(-d^2/(2 \cdot 200^2))$, where d is the distance in km to the nearest district with a riot that year.

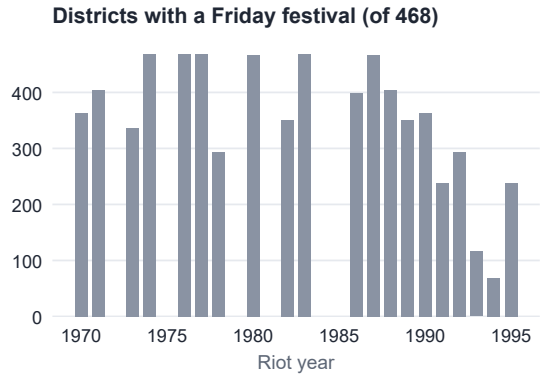
Riot and non-riot district-years

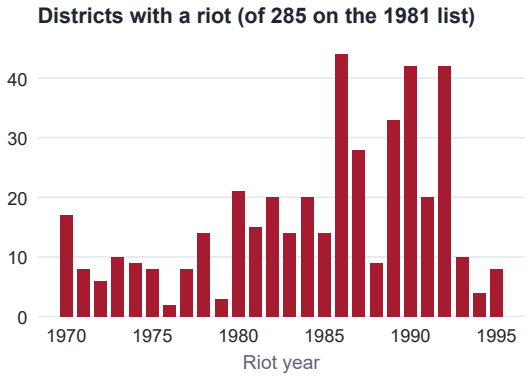
Variable	Panel B: Riot vs. Non-Riot District-Years					
	Riot=1		Riot=0		Difference	P-value
	Mean	SD	Mean	SD		
Outcome Variables						
Muslim Name Index (t+1)	0.750	0.116	0.677	0.151	0.073***	0.000
Hindu Name Index (t+1)	0.789	0.118	0.776	0.075	0.012	0.517
Instrument						
Friday Festival	0.650	0.477	0.569	0.495	0.081***	0.001
Observations	429		6,981			

Notes: District-years of the districts on the 1981 list, the only ones with own-district riot data. Difference: coefficient on riot in a regression of the row variable on riot, with standard errors clustered by district. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Friday festivals and riots by year

Left: districts with a Friday festival (Iyer and Shrivastava's coding) in each riot year. Right: districts on the 1981 list with a riot.





First stage by specification

	Dependent Variable: Riot Measure			
	(1)	(2)	(3)	(4)
Riot Measure	Riot	Riot	Exposure	Count
Festival on Friday	0.0104***	0.0130***	0.0652***	1.0003***
	(0.0031)	(0.0031)	(0.0031)	(0.0539)
	[0.0051]	[0.0048]	[0.0281]	[0.2992]
First-stage F, district clusters	11.46	17.32	435.48	344.87
First-stage F, state-year clusters	4.20	7.28	5.40	11.18
Observations	12,161	12,161	12,161	12,161
R-squared	0.195	0.201	0.304	0.338
Festival Coding	Goyal	Iyer	Iyer	Iyer
Controls	No	Yes	Yes	Yes
Quadratic Year Trend	No	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes

Notes: Riot: a riot in the district, for districts on the 1981 list (0 elsewhere). Exposure: $\exp(-d^2/(2 \cdot 200^2))$, where d is the distance in km to the nearest district with a riot that year. Count: riots in the riot-data districts, weighted the same way. The instrument is 1 if one of the state's three main Hindu festivals, Dushehra or Diwali falls on a Friday, in the coding of Goyal and van Noort (2025) or of Iyer and Shrivastava (2018). Controls: BJP rule, Muslim share, urbanisation, literacy, electricity and tap water. Column 3 is the main specification. Standard errors clustered by district in parentheses and by state-year in brackets; stars use district clusters. $^{***}\,p<0.01$, $^{**}\,p<0.05$, $^{*}\,p<0.1$.

Muslim name index by specification

	Outcome: Muslim Name Index (t+1)			
	(1)	(2)	(3)	(4)
Riot Measure	Riot	Riot	Exposure	Count
Panel A: OLS Estimates				
Riot measure	−0.0032	−0.0033	0.0068***	−0.0007***
	(0.0028)	(0.0026)	(0.0010)	(0.0002)
Panel B: Reduced Form Estimates				
Friday Festival	0.0008	0.0015***	0.0015***	0.0015***
	(0.0006)	(0.0006)	(0.0006)	(0.0006)
	[0.0022]	[0.0020]	[0.0020]	[0.0020]
Panel C: IV Estimates				
Riot measure (instrumented)	0.0766	0.1167**	0.0233***	0.0015***
	(0.0630)	(0.0516)	(0.0085)	(0.0006)
	[0.2148]	[0.1611]	[0.0309]	[0.0021]
First-stage F, district clusters	11.46	17.32	435.48	344.87
First-stage F, state-year clusters	4.20	7.28	5.40	11.18
Observations	12,161	12,161	12,161	12,161
Festival Coding	Goyal	Iyer	Iyer	Iyer
Controls and Quadratic Trend	No	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes

Notes: Columns as in the first stage by specification; column 3 is the main specification. Standard errors clustered by district in parentheses and by state-year in brackets; stars use district clusters. The reduced-form tests are the Anderson–Rubin tests of a zero effect; the last table reports their p-values and 95% sets. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$.

Hindu name index by specification

	Outcome: Hindu Name Index (t+1)			
	(1)	(2)	(3)	(4)
Riot Measure	Riot	Riot	Exposure	Count
Panel A: OLS Estimates				
Riot measure	0.0018***	−0.0003	0.0012**	0.00002
	(0.0007)	(0.0007)	(0.0005)	(0.00004)
Panel B: Reduced Form Estimates				
Friday Festival	0.0009***	0.0014***	0.0014***	0.0014***
	(0.0002)	(0.0002)	(0.0002)	(0.0002)
	[0.0015]	[0.0011]	[0.0011]	[0.0011]
Panel C: IV Estimates				
Riot measure (instrumented)	0.0829**	0.1075***	0.0215***	0.0014***
	(0.0331)	(0.0301)	(0.0033)	(0.0002)
	[0.1437]	[0.0947]	[0.0188]	[0.0012]
First-stage F, district clusters	11.46	17.32	435.48	344.87
First-stage F, state-year clusters	4.20	7.28	5.40	11.18
Observations	12,161	12,161	12,161	12,161
Festival Coding	Goyal	Iyer	Iyer	Iyer
Controls and Quadratic Trend	No	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes

Notes: Columns as in the first stage by specification; column 3 is the main specification. Standard errors clustered by district in parentheses and by state-year in brackets; stars use district clusters. The reduced-form tests are the Anderson–Rubin tests of a zero effect; the last table reports their p-values and 95% sets. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$.

Anderson–Rubin tests by specification

	Anderson–Rubin Tests of a Zero IV Effect			
	(1)	(2)	(3)	(4)
Riot Measure	Riot	Riot	Exposure	Count
Panel A: Muslim Name Index (t+1)				
IV estimate	0.0766	0.1167	0.0233	0.0015
AR p-value, district clusters	0.194	0.007	0.007	0.007
AR p-value, state-year clusters	0.720	0.444	0.444	0.444
AR 95% set, district clusters	[−0.0418, 0.2719]	[0.0311, 0.2711]	[0.0066, 0.0402]	[0.0004, 0.0026]
AR 95% set, state-year clusters	[−0.8886, 2.4251]	[−0.2110, 0.7693]	[−0.0585, 0.1827]	[−0.0026, 0.0077]
Panel B: Hindu Name Index (t+1)				
IV estimate	0.0829	0.1075	0.0215	0.0014
AR p-value, district clusters	<0.001	<0.001	<0.001	<0.001
AR p-value, state-year clusters	0.556	0.197	0.197	0.197
AR 95% set, district clusters	[0.0347, 0.2156]	[0.0645, 0.2120]	[0.0151, 0.0282]	[0.0010, 0.0018]
AR 95% set, state-year clusters	[−0.4215, 2.0950]	[−0.0582, 0.5536]	[−0.0133, 0.1595]	[−0.0008, 0.0051]

Notes: The Anderson–Rubin (AR) test of a zero IV effect is the reduced-form t-test on the Friday festival; the AR 95% set collects the IV effects that the test does not reject. Both stay valid when the instrument is weak, unlike the 2SLS standard errors. Columns as in the first stage by specification; column 3 is the main specification.